

Delta Method Handout

Math 742

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Big Picture

If you can estimate something, the Delta Method tells you how uncertain a function of that estimate is.

As a data scientist, you seldom report raw parameters. What you report is:

- Risk ratios
- Odds ratios
- Log odds
- Probabilities
- AUC differences
- Transformed predictions

→ All of those are non-linear functions of estimators.

→ The delta method allows you to report the standard error of a transformed quantity.

Example: You estimate a probability (i.e. $\hat{p}$). You want to report the log odds:

$$\log\left(\frac{\hat{p}}{1-\hat{p}}\right)$$

How uncertain is the “log-odds”?

Example: A/B testing. Is the new process A, faster than the old process B.

$$\mu_A - \mu_B$$

Often we want the percent change:

$$\frac{\mu_A}{\mu_B}$$

Sometimes it makes sense to take the log of this ratio. Why? Logs are symmetric. Suppose process A is twice as fast,

$$\frac{\mu_A}{\mu_B} = 2$$

What if it's twice as slow?

$$\frac{\mu_A}{\mu_B} = 0.5$$

Not symmetric, but the log is:

$$\log(2) = 0.693, \log(0.5) = -0.693$$

If the original estimator is approximately normal, then a smooth transformation (like a log ratio) is also approximately normal.

The delta method allows you to report the standard error of a transformed (i.e. logged) quantity.

We estimate in log scale because:

- It behaves better statistically.
- It produces cleaner intervals.
- It respects multiplicative structure.

Then we exponentiate and report the percent change.

Delta Method (informal statement)

If

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \sigma^2),$$

then for a smooth function g ,

$$\sqrt{n}(g(\hat{\theta}) - g(\theta)) \xrightarrow{d} N(0, (g'(\theta))^2 \sigma^2).$$

Near the true parameter, every smooth function is approximately linear. The slope of that linear approximation determines how uncertainty propagates.

For the log of the odds ratio:

If $\hat{p}$ has variance $\text{Var}(\hat{p})$, then approximately

$$\text{Var}\left(\log\left(\frac{\hat{p}}{1-\hat{p}}\right)\right) \approx \frac{1}{p^2(1-p)^2} \text{Var}(\hat{p}).$$

If $g(p) = \log(p) - \log(1-p)$; $g'(p)^2 = \frac{1}{p^2(1-p)^2}$